

My model ends at cooling. Real isotope production does not.

I need a public, realistic production timeline so I stop optimizing only the easy half.

PERSONAL NOTE FOR

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My name is Samuel Ogunnubi, and I am a 12th grader in Glen Burnie High School's Biomedical and Allied Health program in Maryland. I started this project after learning what Ac-225 can do in targeted cancer treatment. My original goal was simple: see whether a computer model could help compare production plans faster and eventually help make the process more practical.

THE PROBLEM I FINALLY NOTICED

I built a physics-guided LSTM around irradiation, isotope buildup and decay, and cooling. That is useful for testing the code, but it is not the whole production process. An exact solver is already faster for my small constant-rate equation set. If this project is going to matter, it has to study the real end-to-end timeline and the constraints that actually slow a facility down.

1	2	3	4	5	6	7
Irradiate	Build and decay	Cool	Separate	Purify	Quality check	Recycle or deliver
MODELED	MODELED	MODELED	MISSING	MISSING	MISSING	MISSING

What I already have

- A frozen reduced model for irradiation, decay, and cooling.
- A 10,000-case schedule study with every finalist checked by Radau.
- Produced and recovered activity stored separately in my Berkeley reconstruction.

What I need from real operations

- Public ranges for separation, purification, quality-control, and turnaround time.
- Recovery losses, target reuse, staffing or equipment constraints, and the step that usually controls the schedule.
- One public process example I can cite and reproduce without using private facility data.

My question: Could you recommend a public example that follows an isotope from irradiation through a usable, quality-checked product, or point me to the MURR person who understands that workflow best? I would use it to change my goal from 'predict atoms quickly' to 'compare total time and usable activity under realistic constraints.'

I am not asking for proprietary MURR numbers or a radioactive experiment. A published workflow, reasonable public ranges, or the right person to ask would be enough to move the project forward.